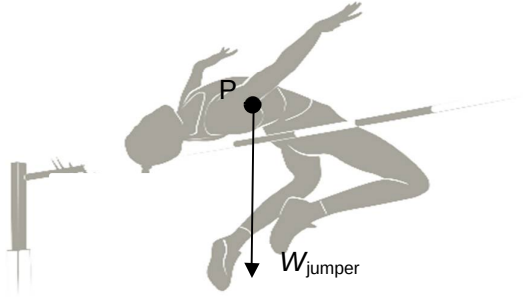
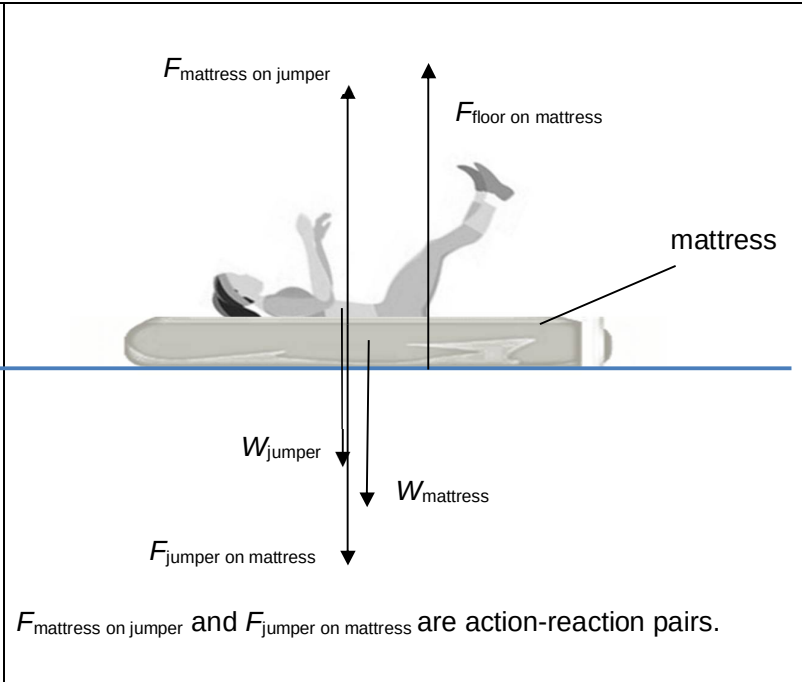


	$\therefore f = 0.0420 \pm 0.0005$	expression
2(a)	Newton's third law of motion states that if body A exerts a force on body B, body B will exert the same type of force of equal magnitude but opposite in direction on body A.	[1]
2(b)(i)		[1] W_{jumper} only

2(b)(ii)	 $F_{\text{mattress on jumper}}$ $F_{\text{floor on mattress}}$ mattress W_{jumper} W_{mattress} $F_{\text{jumper on mattress}}$ $F_{\text{mattress on jumper}}$ and $F_{\text{jumper on mattress}}$ are action-reaction pairs.	[1] for 2 labelled forces on jumper, length of $F_{\text{mattress on jumper}}$ is greater than W_{jumper}, [1] for 3 labelled forces on mattress, length of $F_{\text{floor on mattress}}$ is greatest [1] for showing that $F_{\text{mattress on jumper}}$ and $F_{\text{jumper on mattress}}$ is the action-reaction pair
2(c)(i)	Area under $F - t$ graph gives change in momentum. $ \Delta p = \frac{1}{2}(1.2)(375)$ $ 0 - (50)(u) = 225$ $u = 4.5 \text{ m s}^{-1}$	[1] correct substitution [1] correct answer
2(c)(ii)	By principle of conservation of momentum: $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ $(50)(1.5) + 0 = (50)v_1 + (3.0)v_2 \text{ ---- (1)}$ For elastic collision, relative speed of approach ($u_1 - u_2$) before collision is equal to the relative speed of separation ($v_2 - v_1$) after collision. $u_1 - u_2 = v_2 - v_1$ $1.5 - 0 = v_2 - v_1 \text{ ---- (2)}$ $v_1 = v_2 - 1.5$ Substituting (2) into (1) $(50)(1.5) + 0 = (50)(v_2 - 1.5) + (3.0)v_2$ $v_2 = 2.8 \text{ m s}^{-1}$	[1] correct momentum equation [1] correct substitution [1] correct answer